GENESIS OS / RELATIONSHIP SCORES Operational Closure Specification

Final scoring lock • schema registry • consent and retention • moderation • evidence bands • narrow-v1 rollout

Purpose: Close the remaining operational gaps so engineering can implement the system without guessing, while keeping the original Relationship Scores and Genesis OS architecture intact.

Primary source base	Master Engineering Blueprint, Missing Data Layers Engineering Blueprint, Companion Systems Engineering Guidebook, v4 scoring workbook, and all-on-one scored answer workbook.
What this document locks	Answer-bank scoring appendix, canonical schemas and enums, consent-retention policy, moderation and escalation, evidence bands, and staged launch plan.
Audience	Founders, product, backend, mobile, AI/ML, data, trust and safety, analytics, and QA.
Design rule	Deterministic first. AI may explain, rank, summarize, and propose signals, but may not mutate canonical profile truth without governed evidence.

Document Map

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Important: This specification does not replace the prior blueprints. It closes the unresolved operating gaps and converts them into build-ready contracts, governance rules, and formulas.

1. Executive Closure Summary

The base system is already strong. The unresolved risk is not imagination; it is operational ambiguity. This document turns the remaining fuzzy areas into locked rules. The system can now be built as a governed emotional operating system rather than a stack of brilliant ideas that slowly drift apart in implementation.

- The final answer-bank scoring layer is now locked around the v4 deterministic workbook structure, with a production scoring policy that preserves interpretability while preventing random tuning.
- Every canonical schema, enum, and state key is versioned and managed through a registry contract. No team may invent labels ad hoc in code, prompts, dashboards, or exports.
- Consent, retention, deletion, and derived-data handling are split cleanly so the platform can protect users without destroying auditability or poisoning analytics.
- Moderation now exists as a real system with severity ladders, human-review triggers, rate limits, escalation deadlines, and artifact logging across messaging, ExsAnonymous, Sanctuary, and couple spaces.
- Every forecast and AI insight now carries an evidence band, not just a confidence number, so the product never speaks with fake certainty when the underlying evidence is thin.
- The launch plan narrows v1 intentionally so the engine learns from clean signals before heavy social complexity, couple forecasting depth, grief systems, and advanced economy features expand the blast radius.

2. Non-Negotiable Build Principles

Principle	Production meaning
Deterministic first	All visible scores must be reproducible from stored

Principle	Production meaning
	fields, versioned formulas, and stable lookup tables.
Event-sourced truth	Raw events are never replaced by summaries. Aggregates and AI narratives are layered on top of source truth.
Fast state vs slow identity	Mood, anger, grief, and conversation heat may move quickly; archetype, shadow, and baseline polarity may only drift through governed thresholds.
Canon over synonyms	Stored IDs never change to match marketing language. Human-facing copy may evolve; canonical enums do not.
Consent by scope	A user may consent to one AI scope and deny another. Consent is not a single blanket boolean.
Evidence before certainty	Low-evidence insights must speak softly, and some outputs must stay internal until evidence improves.
Graceful degradation	If AI, pair consent, or secondary services fail, the deterministic core still works.

Hard rule: No companion module may directly overwrite a slow identity trait on its own. It can only contribute evidence to the governor.

3. Final Answer-Bank Scoring Lock

This section locks the production scoring doctrine. It uses the v4 workbook as the deterministic base but closes the remaining ambiguity around row scoring, question-weight bands, penalties, and version control.

3.1 Row-level answer scoring doctrine

- Every answer option remains a multi-axis record, not a single score. One answer row may affect readiness, compatibility, forecast, archetype evidence, shadow evidence, communication style, regulation style, love language, polarity deltas, contradiction pressure, integrity pressure, drift relevance, and forecast relevance.
- The production scoring scale for row quality stays on an eight-point anchored ladder already reflected in the workbook: score_weight in {3,4,5,6,7,8,9,10}; compatibility_weight in {4,5,6,7,8,9,10}; readiness_points and forecast_points in {0,2,4,6,8,10}.
- No answer row may be edited directly in production without a version bump and backfill plan. Answer text edits that materially change psychological meaning require a new answer_id.

3.2 Locked question-weight schedule

Question weights in the v4 workbook already step upward across the test. That schedule is now adopted as production lock because it preserves your design goal: later questions pull more pivotal data. The final schedule is:

Question range	Weight	Reason
Q1-Q8	1.00	Foundational orientation and

Question range	Weight	Reason
		early pattern read.
Q9-Q16	1.05	Slightly higher signal for relational consistency.
Q17-Q26	1.10	Core identity and behavioral tendency.
Q27-Q36	1.15	Higher value because attachment and conflict traits begin concentrating.
Q37-Q40	1.20	Pre-final weighting step for deeper pattern relevance.
Q41-Q44	1.30	High-impact pivotal questions.
Q45-Q48	1.35	Very high-impact late-stage extraction.
Q49-Q50	1.40	Maximum-weight critical questions.
Q51-Q54	1.25	Embedded shadow items; high but slightly below max to prevent shadow domination.
Q55-Q58	1.30	Shadow refinement items.
Q59-Q60	1.35	Highest-weight shadow confirmation items.

Why this stays: It matches the architecture already present in the workbook and avoids the classic mistake of making every question matter equally when the later questions are designed to extract heavier truth.

3.3 Locked user-level readiness formula

Production readiness should use the weighted answer-bank as the anchor, then apply controlled deductions and controlled post-assessment growth deltas. The locked formula is:

ReadinessRaw = 100 * SUM(readiness_points_i * question_weight_i) / SUM(10 * question_weight_i)

ReadinessFinal = clamp(0, 100, ReadinessRaw - ContradictionPenalty - IntegrityPenalty - StabilityDrag + GrowthCredits + RegulationBonus)

- ContradictionPenalty: sum of triggered reflection-cluster penalties, capped at 18 points at user-profile level for any single scoring run.
- IntegrityPenalty: 2 points per active integrity flag in the current assessment, capped at 10 during initial scoring. Later runtime integrity penalties are tracked separately and do not rewrite the original assessment result.
- StabilityDrag: runtime-only deduction applied to current readiness view, not baseline readiness, when fast-state instability or severe moderation events are present.
- GrowthCredits: only earned from validated actions such as Sanctuary modules, relationship exercises, repair completion, or regulation training. Cap at +8 between formal reassessments.
- RegulationBonus: 0 to +4 based on verified regulation behavior, never on self-description alone.

3.4 Locked pair-level compatibility formula

The workbook formula is now adopted as the deterministic production prior because it is interpretable and already aligned with your product logic. The locked compatibility equation is:

CompatibilityScore = clamp(350, 850, 350 + 500 * (0.30 * B + 0.22 * A + 0.18 * (1 - S_risk) + 0.12 * P + 0.08 * Astro + 0.05 * Num) - IntegrityPenaltyPair - ClusterPenaltyPair + InteractionCompatModifier)

Term	Meaning	Locked implementation
B	Behavioral alignment	0.58*BehaviorCore + 0.18*CommNorm + 0.14*RegNorm + 0.10*LoveNorm
A	Archetype chemistry	0.50*PP + 0.20*PS + 0.20*SP + 0.10*SS, normalized from lookup matrix
S_risk	Shadow risk	Shadow pair score normalized and amplified by assessed shadow intensity
P	Polarity dynamics	Remain deterministic; use 0.5 neutral until production polarity table is complete
Astro	Astrology overlay	Lookup score normalized from sign/gender matrix
Num	Numerology overlay	Lookup score normalized from life-path pair table

3.5 Locked relationship forecast formula

Forecast remains pair-level and guidance-oriented. It is not a gate. The locked production formula is:

RelationshipForecast = clamp(0, 100, 100 * (0.30 * RegulationNorm + 0.25 * ConflictRepair + 0.20 * CommunicationHealth + 0.15 * CommitmentPace + 0.10 * PatternStability) - 20 * S_risk - 0.60 * ClusterPenaltyPair + InteractionForecastModifier)

- PatternStability = 0.45 * EmotionalRegulation + 0.25 * ConflictRepair + 0.15 * CommunicationHealth + 0.15 * (1 - S_risk).
- Forecast must always be stored with evidence_band, evidence_score, model_version, scoring_version, and packet_id.

3.6 Locked penalty caps

Penalty / credit lane	Cap	Reason
Assessment contradiction cap	18	Prevents one cluster family from nuking the whole test.
Assessment integrity cap	10	Keeps the engine firm without making it brittle.
Pair cluster penalty cap	35	Matches workbook guardrail for cluster impact.
Pair integrity penalty cap	80	Matches workbook guardrail, but only after pair-level calculation.
Single interaction runtime	20	A single event may not rewrite

Penalty / credit lane	Cap	Reason
penalty		identity; it may only affect fast-state and near-term forecast.
Growth credit cap between reassessments	8	Stops gamified farming.

3.7 Answer-bank governance rules

- No answer row is allowed to have score_weight above 10 or below 3 without a new scoring_version and founder approval.
- If answer text changes by more than cosmetic copy editing, create a new answer_id and deprecate the old row.
- Question number, answer letter, and canonical_q are never repurposed for a different psychological meaning.
- Any change to archetype_id, shadow_type_id, cluster tags, or integrity flags requires a dual review by product psychology owner and scoring owner.
- All answer-bank changes must ship with regression tests against known synthetic profiles and known pair outcomes.

3.8 Regression set required before go-live

Test set	Requirement
Golden singles set	At least 50 synthetic users spanning all 10 archetypes and 6 shadows.
Golden pairs set	At least 150 pair combinations with expected compatibility and forecast band ranges.
Adversarial set	Profiles designed to exploit contradictions, answer gaming, and integrity loopholes.
Drift set	Profiles showing how fast-state changes should and should not alter identity.
Moderation set	Conversations that should trigger safety or abuse moderation without polluting baseline identity.

4. Canonical Schema Registry and Enum Governance

This section turns taxonomy discipline into an operating system. The goal is to prevent service drift, analytics drift, and prompt drift.

4.1 Registry repository structure

- Create a dedicated registry repository or mono-repo folder: /contracts/canonical-registry/.
- Subfolders: /enums, /json-schemas, /event-contracts, /state-vectors, /api-openapi, /lookup-tables, /migration-manifests, /deprecations.
- Every service imports generated contract artifacts from this registry; no service maintains a private copy of enums.
- Every derived row stores taxonomy_version, schema_version, scoring_version, rules_version, and prompt_version when relevant.

4.2 Locked core enums

Domain	Canonical values
archetype_id	ICEBOX, PHANTOM_SEEKER, REGULATED_GROWN_UP, ANALYZER, QUIET_EXIT, FIXER, ROMANTIC_IDEALIST, PERFORMER, SURVIVOR, TRANSLATOR
shadow_type_id	MANIPULATOR, STONEWALLER, LOVE_BOMBER, CHAMELEON, SCOREKEEPER, SELF_SABOTEUR
user_move	WITHDRAW, PRESS_FOR_CLARITY, ESCALATE, APPEASE, REPAIR, JOKE_DEFLECT, UNKNOWN
trigger_label	REJECTION_CUE, AMBIGUITY, DISRESPECT_CUE, CONTROL_LOSS, UNFAIRNESS, SHAME_EXPOSURE, ABANDONMENT_PANIC, BETRAYAL
strategy_class	VALIDATE_THEN_ASK, SLOW_PACE, BOUNDARY_CLARITY, DIRECT_CLARIFICATION, REASSURE_AND_GROUND, MIRROR_AND_SUMMARIZE, PAUSE_AND_SPACE, LIGHTEN_WITH_HUMOR
context_type	daily_checkin, pre_chat, post_chat, post_conflict, post_silence, post_date, post_ritual, conversation_segment, conflict_episode, silence_period, date_event
evidence_band	E0_NONE, E1_THIN, E2_PARTIAL, E3_MODERATE, E4_STRONG, E5_COMPELLING
moderation_severity	S0_NONE, S1_LOW, S2_MEDIUM, S3_HIGH, S4_CRITICAL
consent_scope	ASSESSMENT, MATCHING, MESSAGE_METADATA, MESSAGE_CONTENT_AI, SANCTUARY_AI, EXSANONYMOUS_AI, COUPLE_FORECASTING, RESEARCH_AGGREGATION
record_status	ACTIVE, SUPERSEDED, DEPRECATED, SOFT_DELETED, HARD_DELETED, LEGAL_HOLD

Canonical means canonical: Marketing can call the Regulated Grown-Up something shinier later. The database still stores REGULATED_GROWN_UP.

4.3 State vector contract

- Every state_pre and state_post object must expose the same keys in the same casing.

- Minimum keys: arousal_score, safety_score, clarity_score, repair_openness, attachment_threat, anger_band, anger_pressure, grief_sdi, contradiction_pressure, volatility_score, emotion_code, emotion_family, emotion_cell_id, social_orientation, source_module, confidence, timestamp, window_scope.
- Unknown values are explicit nulls or UNKNOWN enums. Do not silently drop keys.

4.4 Registry change workflow

1. Author opens a registry change proposal with affected enums, schemas, API fields, dashboards, prompts, and migration implications.
2. Contract owner reviews for backward compatibility, duplicate semantics, and whether the concept is genuinely new or just a synonym of an existing label.
3. If accepted, registry version bumps semantically: MAJOR for breaking schema, MINOR for new optional fields, PATCH for descriptions only.
4. Code generation publishes updated typed artifacts to backend, mobile, AI runtime, analytics, and QA fixtures.
5. Migration manifest defines backfill, aliasing, deprecation windows, and rollback strategy.
6. Release is blocked until regression tests pass against contract consumers.

4.5 Enum design rules

- One event may have multiple trigger_labels, but only one user_move.
- Enums use stable machine IDs in screaming snake case; display names live in a separate localization layer.
- No service may infer a new enum on the fly from AI output. AI must map to an existing enum or return UNKNOWN.
- Deprecated enums remain readable for historical rows until a planned archive or remap migration completes.

4.6 Required schemas

Schema	Purpose
answer_row.schema.json	The locked answer-bank row structure.
behavior_event.schema.json	Immutable behavior event payload.
user_emotional_state.schema.json	Daily and contextual self-report or inferred state.
perception_check.schema.json	Perception-vs-reality self-report.
trigger_activation.schema.json	Attachment activation records.
profile_snapshot.schema.json	Versioned identity and growth snapshot.
ai_insight_record.schema.json	Structured AI narrative outputs with evidence metadata.
forecast_record.schema.json	Forecast result with evidence and action bundle.
consent_record.schema.json	Consent decisions by scope.
moderation_case.schema.json	Trust and safety case file.
deletion_job.schema.json	Right-to-delete workflow artifact and completion log.

5. Consent, Retention, Deletion, and Derived-Data Policy

This section is written so product, engineering, legal review, trust and safety, and data can all build from the same source of truth.

5.1 Consent model

- Consent is scoped, explicit, timestamped, reversible, and versioned.
- A single master toggle is not sufficient because users may allow matching but deny message-content AI, or allow Sanctuary AI but deny couple forecasting.
- Consent changes do not rewrite historical audit logs, but they do control future ingestion, future derivation, future display, and retention handling.

Consent scope	What it permits
ASSESSMENT	Score the assessment and store structured profile outputs.
MATCHING	Compute compatibility and show match recommendations.
MESSAGE_METADATA	Use latency, message counts, and silence gaps without reading content.
MESSAGE_CONTENT_AI	Allow content-based labeling, summaries, moderation assist, and insight generation.
SANCTUARY_AI	Allow AI processing in private reflection spaces.
EXSANONYMOUS_AI	Allow AI summarization/pattern extraction in recovery spaces.
COUPLE_FORECASTING	Allow pair-level forecasting using both parties' data.
RESEARCH_AGGREGATION	Allow de-identified aggregate use for model calibration and product analytics.

5.2 Consent records

Table: consent_record

Field	Type	Meaning
id	uuid	Primary key
user_id	uuid	Owner user
scope_id	enum	Consent scope
decision	enum	GRANTED / DENIED / REVOKED
effective_at	timestamptz	When this decision takes effect
expires_at	timestamptz nullable	Optional future expiry
policy_version	text	The terms version shown to user
ui_surface	text	Where consent was collected
evidence_json	jsonb	Screenshot hash, device info, or other proof if required
created_at	timestamptz	Audit

5.3 Revocation rules

- Revocation stops new ingestion for that scope immediately or within a maximum of 15 minutes for queue-based services.
- Future AI derivations for the revoked scope are blocked.
- Existing derived outputs tied exclusively to that scope are hidden from user-facing surfaces after revocation unless legally required to remain visible for a safety case.
- Pair-level forecasts are recomputed when one partner revokes COUPLE_FORECASTING. The pair object remains, but forecast status becomes PARTIAL_CONSENT or DISABLED.
- Historical raw records already collected under valid consent remain in restricted storage until their retention timer or deletion job completes, unless hard deletion is mandated by policy or law.

5.4 Retention schedule

Data class	Default retention	Production rule
Assessment answer rows	Life of account + 24 months after closure	Needed for auditability, rescoring, and user export.
Profile snapshots and score versions	Life of account + 24 months	Supports growth history and support traceability.
Message metadata events	18 months rolling	Useful for behavioral learning without indefinite hoarding.
Message content-derived labels	12 months rolling	Retain structure longer than raw text only if scope remains consented.
Raw message content for AI processing cache	7 days or less	Do not keep long-term unless product surface explicitly stores the content.
Sanctuary AI artifacts	User-controlled archive; default 18 months	Private space requires stronger user control.
ExsAnonymous posts and derived pattern artifacts	User-controlled archive; default 12 months	High sensitivity domain.
Moderation cases	24 months minimum, longer if legal hold	Trust and safety requires longer chain of custody.
Deletion logs	Permanent audit record without personal content	Needed to prove deletion operations.

5.5 Deletion model

- Soft delete hides data from product surfaces and blocks new processing, but does not satisfy a full deletion request.
- Hard delete removes the identifiable source rows and any directly linked derived rows not subject to legal hold.
- De-identification removes or irreversibly transforms personal identifiers while preserving aggregate statistics if the user granted RESEARCH_AGGREGATION or policy allows lawful aggregate retention.
- Deletion jobs must be asynchronous, idempotent, auditable, and restart-safe.

5.6 Derived-data handling

Derived-data class	Policy
Category A - directly derived and user-identifiable	Delete when source is hard-deleted or consent scope requires purge.
Category B - pair-level outputs referencing a deleted user	Delete or tombstone pair outputs; surviving user sees pair as unavailable.
Category C - de-identified aggregate metrics	Retain if no personal re-identification risk and policy permits aggregate analytics retention.
Category D - moderation evidence on legal hold	Retain under restricted access until hold release.

Do not fake deletion: If a score, summary, or forecast is still reconstructable to a deleted person, it is not deleted enough.

5.7 User export rights

- Users may export assessment answers, score history, profile snapshots, stored reflections, consent history, and a plain-language explanation of active model versions affecting their outputs.
- Exports should not include raw other-party private data, hidden moderation heuristics, or partner-specific confidential notes.
- Every export file must show scoring_version, taxonomy_version, and export_generated_at.

6. Moderation and Escalation Operating Model

The product is emotionally intimate. That means moderation is not optional plumbing; it is part of core system safety.

6.1 Coverage surfaces

- Direct messaging
- Knocks and introductions
- Couple mode shared spaces
- Whisper Window if sent to another user
- ExsAnonymous posts, comments, and partner autopsy tools
- Sanctuary content only when user has consented to AI processing or when explicit safety emergencies are detected under policy
- Profile text, prompts, and uploaded user-facing media where supported

6.2 Severity ladder

Severity	Primary action	Typical triggers
S0_NONE	No action	Normal content or benign conflict.
S1_LOW	Log + passive monitor	Mild rudeness, repeated sarcasm, low-grade disrespect.
S2_MEDIUM	Warning, friction prompts, rate limits, or temporary feature cooling	Harassment patterns, coercive pressure, repeated manipulation markers, sexual pressure, repeated boundary pushing.
S3_HIGH	Immediate hide, lock, human review queue, partner protection steps	Threats, hate, intimidation, doxxing attempt, revenge content, severe harassment, credible coercive control markers.
S4_CRITICAL	Emergency workflow, human escalation, safety resources, account freeze	Self-harm threat, credible violence threat, child safety issue, non-consensual intimate image risk, active stalking signal.

6.3 Required moderation event schema

Field	Type	Meaning
id	uuid	Primary key
subject_type	enum	message, post, profile, conversation, relationship, user
subject_id	uuid	Target artifact
detector_type	enum	rule_engine, ai_classifier, user_report, moderator_action
risk_categories	text[]	harassment, coercion, self_harm, threat, hate, sexual_pressure, privacy, fraud, etc.
severity	enum	S0_NONE to S4_CRITICAL
confidence_score	numeric(5,2)	0 to 1
evidence_refs	jsonb	Pointers to source artifacts

Field	Type	Meaning
status	enum	OPEN, QUEUED, ACTIONED, APPEALED, CLOSED
created_at	timestamptz	Audit

6.4 Escalation rules

1. Any S4_CRITICAL detection immediately creates a moderation_case, freezes risky outbound surfaces as configured, and places the case into a priority human queue.
2. S3_HIGH creates a moderation_case within seconds and may hide content immediately pending review.
3. S2_MEDIUM may auto-warn, down-rank, or rate-limit while also contributing to a rolling behavior risk score.
4. Three S2 events in 30 days or one S3 event moves the account into heightened observation mode.
5. Repeated manipulation, coercive control, or boundary-violation patterns may reduce messaging privileges even when each individual event is below removal threshold.

6.5 Human review service levels

Queue type	Target response	Note
S4_CRITICAL	Immediate to 15 min	Emergency path
S3_HIGH	Within 1 hour	High risk
S2_MEDIUM appeals	Within 24 hours	Standard review
S1_LOW	No mandatory human review unless pattern escalates	Automated logging sufficient

6.6 Specific high-risk categories and product rules

Self-harm or suicide content: Never return purely relational coaching. Show crisis-support routing, create a critical case, and suppress romantic or gamified surfaces until risk workflow completes.

Threats of violence: Freeze outbound communication surface, preserve evidence, and escalate to human review immediately.

Abuse or coercive control: Reduce trust in pair-health insights, protect the potentially targeted user, and avoid giving the aggressor tactical guidance on how to argue better.

Revenge or humiliation content in breakup spaces: Auto-hide pending review; do not let ExsAnonymous become Yelp for trauma with worse lighting.

Privacy violations and doxxing: Immediate removal and case creation.

Child safety or explicit exploitation: Immediate critical path and policy-defined reporting steps.

6.7 Moderation + scoring separation

- Moderation events may affect runtime trust tier, messaging rights, and near-term forecast confidence.
- Moderation events do not directly rewrite archetype or baseline readiness without governed long-window evidence.
- A severe moderation event may add a temporary safety overlay to pair outputs, but the underlying historical scores stay auditable.

7. Evidence Bands for Forecasts and AI Insights

Confidence is not enough. A model can be numerically confident while still relying on thin, one-sided, or stale evidence. Every meaningful insight must therefore carry an evidence band.

7.1 Evidence score model

EvidenceScore = 0.30 * SourceCoverage + 0.20 * RecencyStrength + 0.20 * ObjectiveSignalDepth + 0.15 * SubjectiveSignalDepth + 0.10 * CrossSourceAgreement + 0.05 * ConsentCompleteness

- SourceCoverage measures how many relevant streams contribute: assessment, messaging metadata, message-content AI, Sanctuary, ExsAnonymous, couple reflections, and prior outcome history.
- RecencyStrength measures how fresh the signals are for the output window.
- ObjectiveSignalDepth measures count and quality of behavioral events, not just self-report.
- SubjectiveSignalDepth measures presence of check-ins, perception checks, trigger logs, and contextual notes.
- CrossSourceAgreement rewards coherence and penalizes contradiction.
- ConsentCompleteness prevents the product from speaking like it sees both sides when it only has one.

7.2 Locked evidence bands

Band	Score range	Display rule
E0_NONE	0.00-0.09	Do not show user-facing insight. Store internal placeholder only.
E1_THIN	0.10-0.29	Show only soft language such as 'early signal' or keep internal by product choice.
E2_PARTIAL	0.30-0.49	Show guidance with caution phrasing; never strong causal language.
E3_MODERATE	0.50-0.69	User-facing summary allowed; action suggestions may be shown.
E4_STRONG	0.70-0.84	Strong but still non-diagnostic language allowed.
E5_COMPELLING	0.85-1.00	High-confidence, multi-source evidence with strong freshness and agreement.

7.3 Forecast display rules by band

Band	What user sees
E1_THIN	Do not surface numeric forecast to users; internal only or 'not enough shared data yet'.
E2_PARTIAL	Show forecast band label only, no exact number, plus note that more shared data improves accuracy.
E3_MODERATE	Show exact forecast and top 2 drivers.
E4_STRONG	Show exact forecast, top drivers, and recommended interventions.
E5_COMPELLING	Show exact forecast, drivers, evidence explanation,

Band	What user sees
	and trend comparison.

7.4 AI narrative language policy

- E1/E2 outputs use language like: 'early signal,' 'possible pattern,' 'based on limited evidence,' or 'this may change as more interaction data arrives.'
- E3 outputs may say: 'current pattern suggests.'
- E4/E5 outputs may say: 'the available evidence strongly suggests,' but may never use diagnostic language unless the product is explicitly designed for licensed clinical use, which it is not.
- If ConsentCompleteness < 0.50, all pair insights must explicitly say they are based on partial shared visibility.

7.5 AI insight record contract additions

New field	Type	Meaning
evidence_score	numeric(5,2)	0 to 1
evidence_band	enum	E0_NONE to E5_COMPELLING
evidence_breakdown_json	jsonb	Per-component contribution and coverage
consent_completeness	numeric(5,2)	0 to 1
display_policy	enum	INTERNAL_ONLY, SOFT_USER_VISIBLE, FULL_USER_VISIBLE
band_reason_text	text	Plain-language explanation

8. API Contracts and Service Boundaries

This section defines the minimum contracts required so teams can build the system without silently diverging.

8.1 Core services

Service	Responsibility
assessment-service	Stores answer selections, computes readiness baseline, archetype/shadow vectors, and emits profile snapshots.
pairing-service	Computes compatibility, forecast prior, access tier, and pair summary.
event-ingest-service	Accepts messaging metadata, content labels, silence events, and contextual behavior events.
state-service	Stores user_emotional_state, trigger_activation, perception_check, and state vectors.
governor-service	Applies drift rules, slow/medium/fast write lanes, and evidence-based update decisions.
moderation-service	Runs automated detectors, case management, action execution, and appeals.
consent-service	Evaluates scope permission on every processing request.
deletion-service	Coordinates purge, tombstone, and audit workflows.
insight-service	Generates AI narrative records from normalized emotional data packets.
analytics-export-service	Pushes approved de-identified aggregates to warehouse.

8.2 Required API endpoints

Endpoint	Contract purpose
POST /v1/assessments/{user_id}/answers	Upsert answer selections by answer_id
POST /v1/assessments/{user_id}/score	Run deterministic assessment scoring and snapshot creation
POST /v1/pairs/score	Compute compatibility and forecast prior for two users
POST /v1/events/behavior	Ingest immutable behavior_event payload
POST /v1/states/checkin	Create user_emotional_state record
POST /v1/perception-checks	Create perception_check record
POST /v1/triggers/activation	Create trigger_activation record
POST /v1/insights/generate	Create ai_insight_record from packet_id
POST /v1/moderation/report	User report intake
POST /v1/consent	Grant, deny, or revoke scope
POST /v1/deletion-jobs	Open deletion workflow
GET /v1/users/{id}/export	User data export bundle

8.3 Example emotional data packet

```
{
  "packet_id": "uuid",
  "subject_type": "pair",
  "subject_id": "uuid",
  "window_days": 30,
  "taxonomy_version": "1.0.0",
  "scoring_version": "v4.1-lock",
  "consent_completeness": 0.74,
  "evidence_score": 0.63,
  "assessment": {
    "user_a": {"readiness_score": 81.4, "primary_archetype_id": "REGULATED_GROWN_UP"},
    "user_b": {"readiness_score": 67.2, "primary_archetype_id": "ANALYZER"}
  },
  "behavior_rollup": {
    "initiator_ratio_a": 0.58,
    "repair_latency_hours": 7.2,
    "silence_gap_count_30d": 3,
    "anger_r4_plus_events_30d": 1
  },
  "subjective_rollup": {
    "avg_felt_safety": 6.8,
    "avg_felt_heard": 6.2,
    "self_awareness_delta": 1.1
  },
  "forecast_inputs": {
    "communication_health": 0.72,
    "conflict_repair": 0.61,
    "commitment_pace": 0.70,
    "pattern_stability": 0.64
  }
}
```


9. Failure Handling and Edge-Case Control

Scenario	Required behavior
AI service unavailable	Proceed with deterministic scoring and store ai_status = DEGRADED. Do not block core app flows.
One partner denies couple forecasting	Allow messaging or relationship mode if product rules permit, but pair forecast becomes PARTIAL_CONSENT or hidden.
Message content consent revoked mid-thread	Stop content labeling immediately; metadata processing continues only if separately consented.
Schema version mismatch between services	Reject write and raise contract error. Silent coercion is forbidden.
Deletion job collides with moderation legal hold	Apply hold precedence, purge what is allowed, and issue partial completion record.
Evidence band below threshold for UI surface	Store internal output only and return user-friendly insufficiency state.
Conflicting AI and deterministic labels	Deterministic canonical label wins unless review process explicitly remaps logic in a future rules version.
Assessment retake while pair forecasts active	Create new scoring snapshot; pair service recomputes using latest active snapshot versions without overwriting historical pair runs.

10. Narrow-v1 Launch Plan So the Engine Learns Before the Empire Expands

The smartest launch is not the biggest launch. It is the cleanest one. v1 should maximize high-quality signal collection while minimizing social complexity, moderation load, and trust risk.

10.1 What v1 should include

- Onboarding and assessment engine
- Individual readiness score and archetype/shadow outputs
- Compatibility scoring using deterministic prior
- Knock system and controlled direct messaging
- Message metadata event logging
- Optional message-content AI only for users who grant that scope
- Sanctuary Lite: check-ins, reflections, emotional mirror, simple regulation exercises
- Consent center, export, and deletion request surfaces
- Moderation service from day one

10.2 What v1 should not include yet

- Deep couple-mode forecasting based on thin interaction history
- Full ExsAnonymous social graph at scale
- Aggressive coin economy loops that encourage farming rather than truth
- Public scoreboards with emotionally sensitive comparisons
- Too many companion systems writing into baseline identity simultaneously
- High-complexity licensing APIs exposed to third parties before internal learning stabilizes

10.3 v1 learning goals

Learning target	Why it matters
Assessment integrity	Which questions separate archetypes cleanly and which create noise?
Pair signal quality	Which deterministic drivers best predict early conversation success, ghosting, repair, or drop-off?
Moderation load	What categories trigger most often and where do false positives live?
Consent friction	Which scopes users accept or deny, and where explanation copy needs improvement?
Evidence calibration	How quickly can the system move from E1/E2 to E3+ for pair forecasts?
Growth mechanics	Which validated actions actually correlate with better repair and lower volatility?

10.4 Launch phasing

Phase	Scope
v1.0	Assessment, deterministic compatibility, Knock, controlled messaging, metadata events, moderation, Sanctuary Lite.
v1.1	Perception checks, trigger logs, evidence bands on user-facing forecasts, stronger export and deletion tooling.
v1.2	Selective pair forecasting depth, richer state logging, initial governor rollups, couple exercises.
v1.5	Companion systems such as ExsAnonymous and RS games feed evidence through caps and drift gates.
v2.0	Deeper couple mode, broader forecasting, licensing surfaces, advanced social systems after calibration.

10.5 Key launch metrics

- Assessment completion rate
- Consent-by-scope opt-in rates
- Knock acceptance rate
- Conversation survival at day 3, 7, 14, and 30
- Repair event frequency and repair latency
- Moderation rate per 1,000 conversations
- Forecast evidence-band progression
- Forecast calibration error by band
- Deletion-job completion SLA
- False-positive and false-negative moderation review rate

Practical launch truth: v1 wins by being believable, safe, and measurable. It does not need to show every trick in the castle on day one.

Appendix A - Locked Answer-Row Schema

Field	Type	Meaning
answer_id	string	Immutable row key such as Q01_A.
canonical_q	int	Canonical question number.
question_weight	numeric(4,2)	Locked weight schedule.
answer_text	text	Human-visible answer.
score_weight	smallint	3 to 10 ladder.
compatibility_weight	smallint	4 to 10 ladder.
readiness_points	smallint	0,2,4,6,8,10.
forecast_points	smallint	0,2,4,6,8,10.
archetype_id	enum	Primary archetype evidence.
shadow_type_id	enum nullable	Shadow evidence if present.
comm_style	enum	Communication style mapping.
regulation_style	enum	Regulation style mapping.
love_language	enum	Love language mapping.
cluster_tags	text[]	Reflection cluster links.
integrity_flag	boolean	Assessment integrity marker.
drift_relevance	boolean	Whether row matters for drift analysis.
forecast_relevance	boolean	Whether row matters for forecast features.
version_snapshot	text	Scoring/version label.

Appendix B - Minimum SQL Tables Added by This Closure Spec

Table	Purpose
consent_record	Consent by scope
moderation_event	Automated or user-reported risk detections
moderation_case	Human-review case management object
deletion_job	Delete / de-identify workflow coordinator
schema_registry_audit	Registry change ledger
pair_output_visibility	Stores whether pair outputs are hidden due to consent, evidence, or moderation
evidence_breakdown	Optional normalized component storage if not embedded in insight/forecast JSON

Appendix C - Required Review Boards

Board	Minimum members
Scoring Review Board	Founder/product, scoring owner, data owner, QA
Contract Review Board	Backend lead, mobile lead, data lead, AI lead
Trust and Safety Board	Moderation lead, product, support, policy owner
Deletion and Privacy Board	Backend, legal/policy, security, analytics owner

Appendix D - Pre-Go-Live Checklist

- All canonical enums published from registry and imported by every service
- Golden synthetic regression suite passes
- Evidence bands displayed correctly in API and UI
- Consent revocation tested end-to-end
- Deletion workflows tested for normal, partial, and legal-hold scenarios
- Moderation critical path tested with human queue and audit trail
- Assessment rescoring versioned and historical snapshots preserved
- Pair outputs recompute correctly after new assessment versions
- Analytics excludes hard-deleted identities and honors de-identification rules
- Fallback behavior verified for AI downtime

Appendix E - Build Order

1. Lock answer-bank CSV/XLSX as scoring_version v4.1-lock and publish it to the registry.
2. Ship contract artifacts and typed models to every service.
3. Implement consent-service and gate all processing through it.
4. Implement deterministic assessment-service and pairing-service first.
5. Implement moderation-service before broad messaging rollout.
6. Add evidence-band calculation to forecast and insight pipelines.
7. Add deletion-service and export-service before opening public beta.
8. Expand to richer state logging and companion systems only after v1 telemetry is healthy.

Appendix F - Final Word to Developers

Build the engine so it can explain itself. This system will live or die on trust. The easiest way to lose trust is to let one service invent its own truth, one prompt invent a new label, or one flashy feature speak more confidently than the evidence allows. Keep the source-of-truth chain clean, keep the registry strict, and let the learning happen in layers.